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Redefining soil health and food security through tropical conservation agriculture

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Conservation agriculture (CA) is a leading strategy for promoting soil health, food production, and ecological sustainability. Yet, food security in the tropics remains elusive, constrained by persistent soil degradation. This systematic review of 474 studies evaluates the state, gaps, and prospects of CA and soil health research across the tropics. Brazil accounted for 40% of the research output, driven by strong institutional support and public investment. Only 17% of studies integrated chemical, physical, and biological soil health indicators, and even fewer applied structured soil health indices. Most research focused on the top 0–20 cm of Oxisols, Alfisols, and Ultisols, limiting the evidence of CA for long-term sustainability. CA systems were dominated by maize and soybean, with low inclusion of root and tuber crops essential for tropical food security. Scaling the Brazilian CA model through South -South cooperation offers a pathway to more resilient and equitable food systems in the tropics.

Conservation agriculture (CA) is a landmark agricultural technology that is globally promoted to improve soil health and ensure sustainable crop production and environmental protection[1]. Originating in the United States during the 1930s as "conservation tillage," CA initially focused on maintaining sufficient soil cover with crop residues to minimize erosion[2]. Over time, this concept evolved into a holistic framework defined by three core principles: minimal soil disturbance, permanent soil cover, and crop diversification[3]. These principles of CA promote a synergistic interplay between soil and plant systems, enabling broad adaptation across diverse agroecological systems. Globally, an estimated 205 million hectares are under Conservation agricultural practices, representing 14.7% of global cropland, with an annual increase of 10 million hectares since 2008[4]. This widespread adoption reflects its recognized benefits for soil conservation, productivity, and resilience.

The technical implementation of CA is closely linked to improved soil health, defined as the soil's capacity to continue functioning as a vital living ecosystem that supports plants, animals, and humans[5]. Healthy soils are pivotal in achieving several Sustainable Development Goals (SDGs), including zero hunger (SDG 2), good health and well-being (SDG 3), and climate action (SDG 13)[6]. This underscores that strengthening soil health through sustainable agricultural practices such as CA is central to long-term food system resilience and environmental sustainability[7–9].

Tropical regions require urgent and targeted agricultural transformation. Despite abundant natural resources and vast arable land, the tropical region continues to underperform in meeting the food security demands of its burgeoning populations[10,11]. Many tropical countries remain net food importers, underscoring a persistent gap between agricultural potential and productivity. This underperformance is largely driven by the dominance of deeply weathered, nutrient-depleted, and erosion-prone soils[12,13]. Additionally, unsustainable land use practices, poor soil management, accelerating impacts of climate change, and population pressure have contributed to the widespread soil degradation[14–16]. Addressing these challenges requires coordinated strategies that support the adoption of CA strategies, including the development of standardized field and laboratory protocols, the establishment of comprehensive soil health databases, implementation of impactful educational programs, and the creation of enabling policy frameworks[17,18]. Crucially, these efforts must account for the unique ecological, cultural, and socioeconomic complexities of tropical landscapes to ensure context-relevant soil health interventions.

Although previous studies have reported positive impacts of conservation strategies on soil conditions in tropical regions, most findings are based on agronomic trials that evaluate changes in soil health indicators[19–22]. To date, there has been no comprehensive, systematized assessment of CA on soil health across the diversity of tropical agroecosystems. Such a synthesis is essential to answer critical questions: What challenges hinder CA adoption in the tropics? What soil health opportunities exist that can be leveraged? Where are the major research and application gaps? Can adopting CA improve food production?

To answer these research questions, we conducted a systematic evaluation of peer-reviewed literature focused on soil health responses to CA in tropical environments. Our objectives were to: (i) map the past 50 years of CA research in the tropics by analyzing temporal and geographic trends in CA research, cataloging the soil health indicators and soil sampling depth being used to assess soil health response to CA, and identifying crop diversity; (ii) quantify the impact of CA research on soil health outcomes; and (iii) explore opportunities in the tropics. Through this approach, we aim

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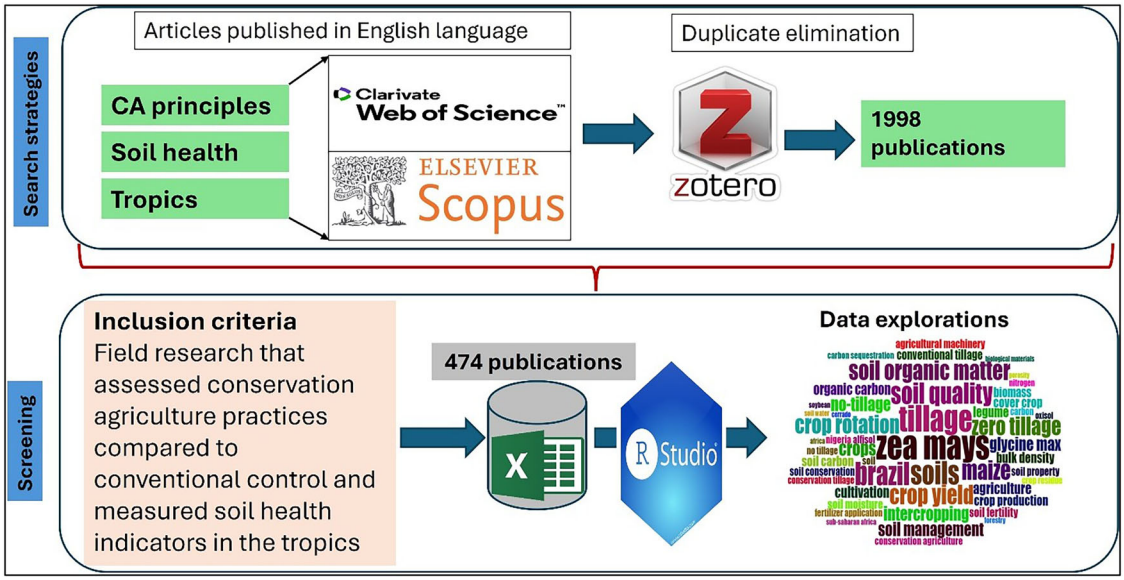


Fig. 1 | Flowchart and design of the systematic review process on conservation agricultural research on soil health in the tropics.

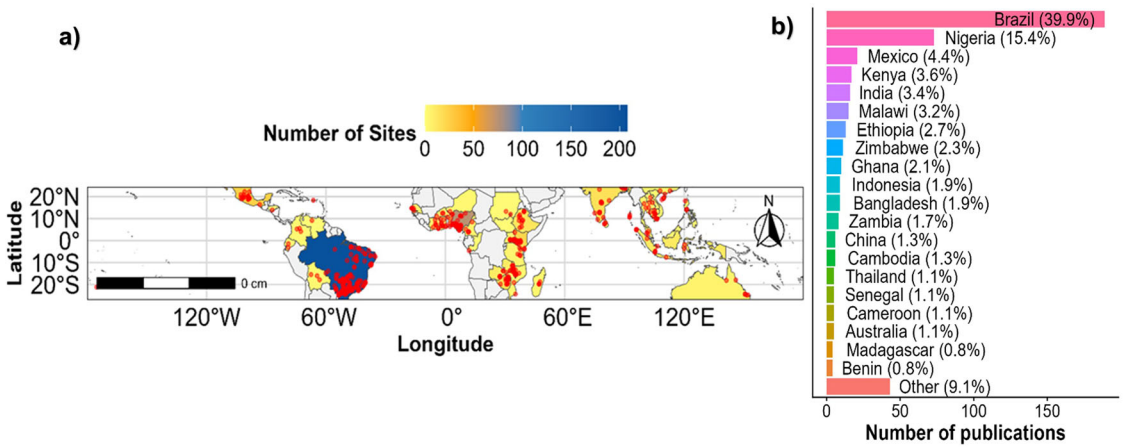


Fig. 2 | Distribution of study locations within the tropics. a Distribution of georeferenced study sites used in the review. **b** contributions per country.

to highlight critical knowledge gaps, identify research opportunities, and guide future studies at the intersection of CA and soil health, ultimately improving soil and food security in the tropics. Ultimately, our findings are intended to support evidence-based policymaking, strategic investments, and region-specific innovations that enhance sustainable agriculture and food security across the tropics.

Results

Bibliometric analysis

A total of 474 studies met the inclusion criteria (Fig. 1; Methods), with about 40% from Brazil (Fig. 2). The bibliometric analysis of the 474 studies revealed an average 7% annual growth in research on CA and soil health in tropical regions. From that total, 262 (55.4%) papers were published in the last decade (Fig. 3a). The earliest in this review was published by Ratan Lal in 1974, which studied the impact of zero versus conventional tillage (disc plow plus harrow) on soil properties at the International Institute of Tropical Agriculture (IITA), Ibadan, Nigeria. Rattan Lal was the most prolific author, contributing 19% of the total authors and maintaining consistency for over four decades, while recent contributors have also gained prominence in CA research (Fig. 3b). Institutional analysis revealed that Brazilian institutions dominated the top positions, with strong representation from Europe and the United States (Fig. 3c and Supplementary Table S1). Conversely, institutions from tropical Africa were notably underrepresented among the top 50 contributors (Supplementary Table S1). This institutional dominance, aligned with the primary funding sources, which were dominated by Brazil, led by CNPQ [National Council for Scientific and Technological Development (Portuguese: Conselho Nacional de Desenvolvimento Científico e Tecnológico)] and CAPES [Brazilian Federal Agency for Support and Evaluation of Graduate Education (Portuguese: Coordenação de Aperfeiçoamento de Pessoal de Nível Superior)] (Fig. 3d and Supplementary Table S2). The strongest international collaboration was between Brazil and the USA (Fig. 3e). Tropical countries substantially collaborated with European nations, particularly Germany, France, the United Kingdom, and Belgium, while intra-tropical collaborations were few (Fig. 3e).

Measured soil health attributes

There is a strong emphasis on chemical and physical soil properties used as soil health indicators within the tropics. Out of the total, 73% studies quantified the impact of CA on at least one soil chemical property (e.g., pH, and/or EC, and or plant nutrients), and 66% measured soil physical properties, while only 33% of the studies measured biological soil health indicators (Fig. 4a). Only 17% of studies integrated all three indicators, and 11 papers assessed the soil health index (Supplementary Table S3). Physical and chemical properties were assessed since the 1990s, while biological indicators became increasingly important after the year 2000 (Fig. 4b). Soil organic

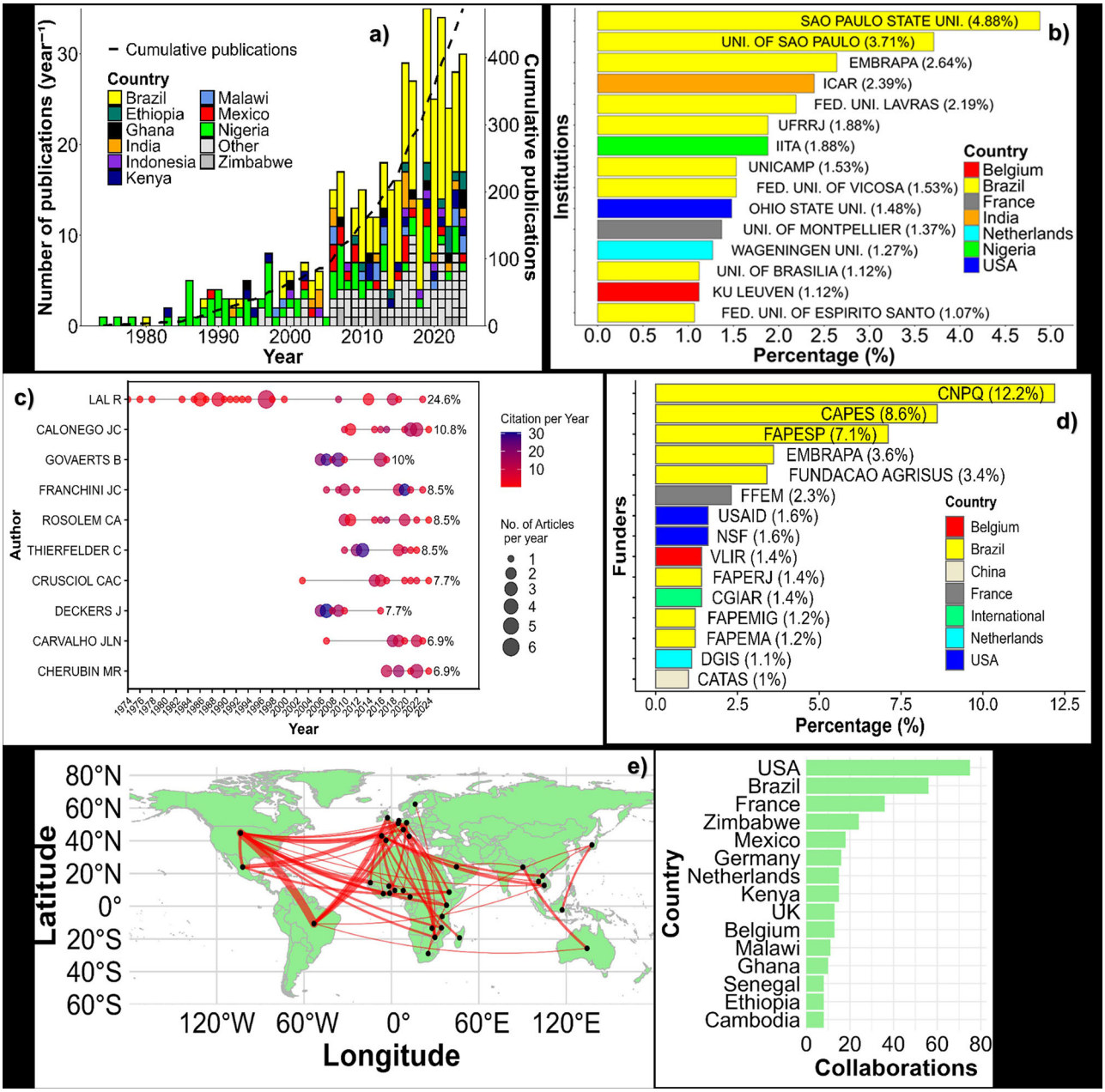


Fig. 3 | Conservation agriculture and soil health research in the tropics. a annual and cumulative number of studies. **b** top 15 institutional contributions. **c** top 10 prolific authors. **d** top 15 highest funding agencies. **e** research collaboration network. Institutions: São Paulo State University (SAO PAULO STATE UNI.), University of São Paulo (UNI. OF SAO PAULO), Empresa Brasileira de Pesquisa Agropecuária (EMBRAPA), Central Research Institute for Dryland Agriculture (ICAR), Universidade Federal De Lavras (FED. UNI. LAVRAS), International Institute of Tropical Agriculture (IITA), Universidade Federal Rural do Rio de Janeiro (UFRRJ), Universidade Federal de Viçosa (FED. UNI. OF VICOSA), Universidade Estadual de Campinas (UNICAMP), Ohio State University (OHIO STATE UNI.), University of Montpellier (UNI. OF MONTPELLIER), Wageningen University (WAGENINGEN UNI.), Katholieke Universiteit Leuven (KU LEUVEN), Universidade de Brasília (UNI. OF BRASILIA), Universidade Federal do Espírito Santo (FED. UNI. OF ESPIRITO SANTO). Funders: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPQ), Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Fundação de Amparo À Pesquisa do Estado de São Paulo (FAPESP), Empresa Brasileira de Pesquisa Agropecuária (EMBRAPA), Agence Française de Développement (FFEM), National Science Foundation (NSF), United States Agency for International Development (USAID), Consortium of International Agricultural Research Centers (CGIAR), Fundação Carlos Chagas Filho de Amparo À Pesquisa do Estado do Rio de Janeiro (FAPERJ), Flemish Inter-University Council (VLIR), Fapemamaranho Research Foundation (FAPEMA), Fundação de Amparo À Pesquisa do Estado de Minas Gerais (FAPEMIG), Dutch Ministry For International Cooperation (DGIS), Central Public-Interest Scientific Institution Basal Research Fund For Chinese Academy Of Tropical Agricultural Sciences (CATAS).

carbon was the most measured parameter across all studies (Fig. 4c). Other frequently assessed chemical indicators included nitrogen, phosphorus, potassium, and pH, while bulk density and soil moisture were the most reported physical indicators. Biological measurements were comparatively limited, with microbial biomass and respiration being the most measured metrics.

Inherent soil properties

Inherent soil properties strongly influenced the impact of CA. The majority of studies were conducted on Oxisols (36.5%), Alfisols (26.8%), and Ultisols (18.2%), with the remaining 18.5% focused on Entisols, Inceptisols, Vertisols, Mollisols, Andisols, and Histosols (Fig. 5a). The reviewed studies cover all United States Department of Agriculture soil

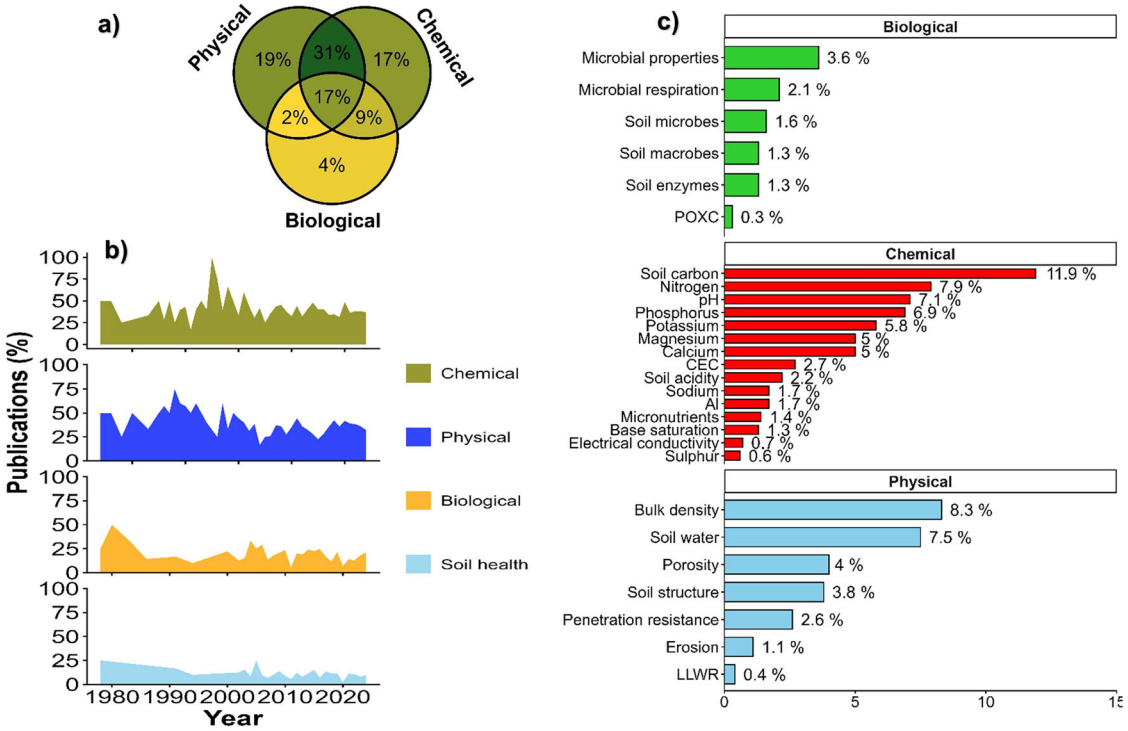
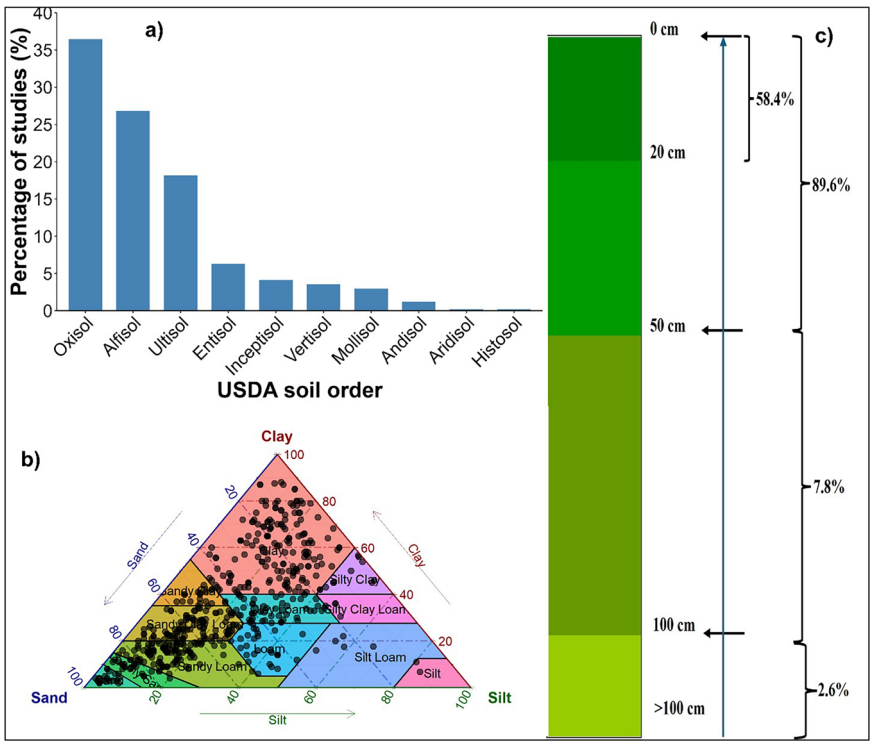


Fig. 4 | Soil health measurements in the tropics. a Annual proportion. **b** Venn Diagram for occurrences and intersections. **c** distribution of soil health related to physical, chemical, and biological indicators. Cation exchange capacity (CEC), Aluminium (Al), and least limiting water range (LLWR).

Fig. 5 | Inherent factors driving soil health in the studies. a Soil orders. **b** soil textural distribution. **c** soil depths assessed in research on conservation agriculture and soil health in the tropics.



textural classes, as presented in the soil textural triangle (Fig. 5b). The most studied textures were clay (31.9%), sandy loam (20.5%), and sandy clay loam (19.6%) (Supplementary Table S4). Soil sampling primarily focused on the top 0–20 cm layer (58.4%), and 89.6% evaluated to soil depth of 0-50 cm (Fig. 5c). Only 2.6% of studies assessed depths beyond 100 cm.

Productions in conservation agriculture

This study analyzed the distribution of crop components integrated into CA systems across tropical regions (Fig. 6). Pastures and cover crops were the most represented category, accounting for 40.1%. Cereals constituted 23.3% of the crop group, with the predominance of *Zea mays* (maize) (Fig. 6). Legume crops constituted 19.9%, dominated by soybean (51.3%) and *Vigna*

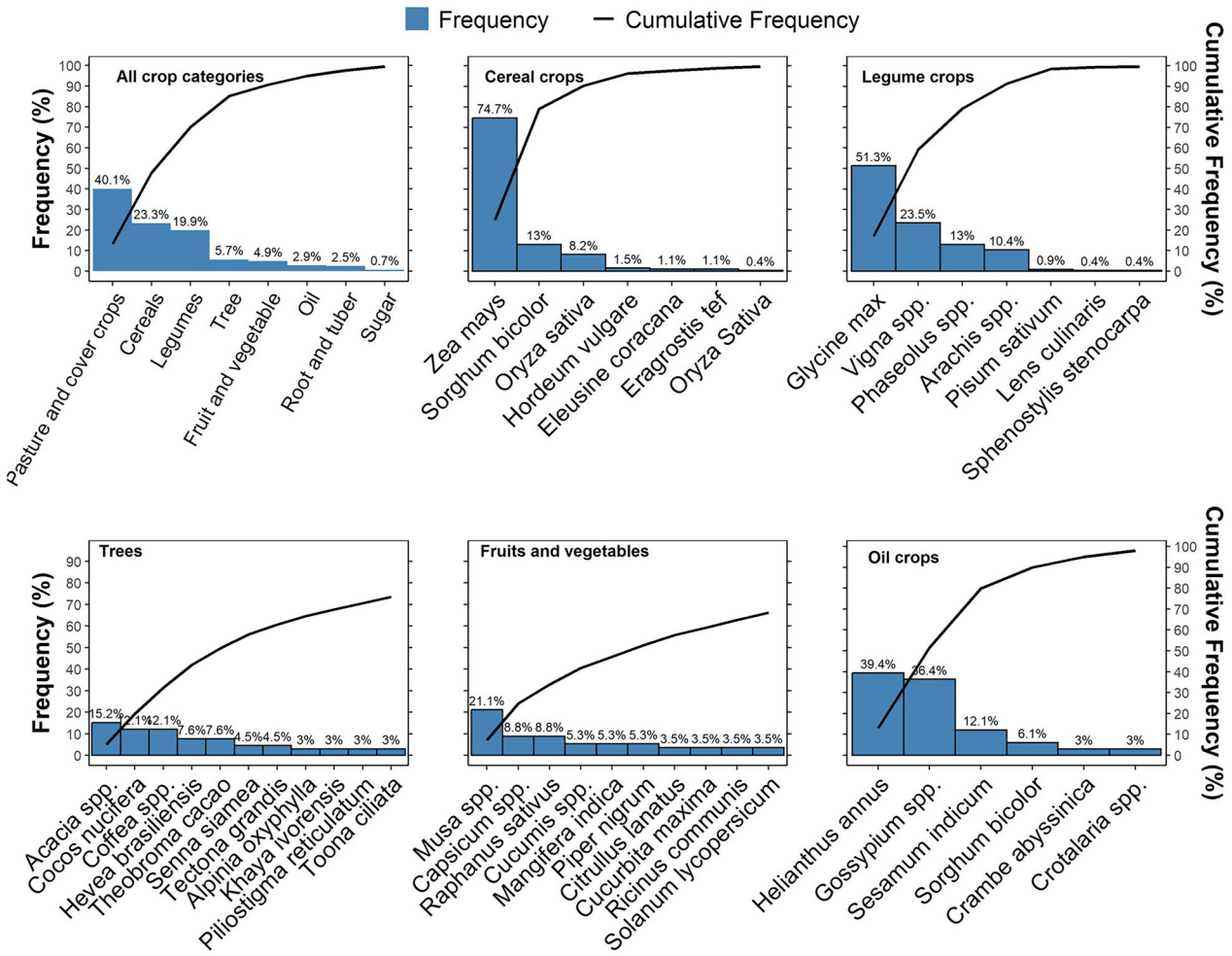


Fig. 6 | Distribution (frequency and cumulative frequency) of crops adapted in conservation agriculture in the tropics.

spp. (23.5%). Trees were included in 5.7% of studies, primarily *Acacia spp*. and *Cocos nucifera*. Fruits and vegetables accounted for 4.9%, with *Musa spp*. been the most common. Oil crops (2.5%), root and tuber crops (2.5%), and sugar crops (0.7%) were the least represented. Most of the research management practice was conducted under mechanized compared to manual farming practices (Fig. 7). There was higher assessment of soil health under zero and conventional tillage, and predominantly associated with mechanized systems.

Soil health under conservation agriculture and food security in the tropics

The distribution of CA research across tropical countries is misaligned with the severity of food insecurity. Most countries with high Global Hunger Index (GHI) scores, particularly in sub-Saharan Africa, had low CA research output (Fig. 8). Whereas Brazil, which contributed the highest research output, had a low hunger score despite the high population, while other highly populated countries, such as Nigeria, Indonesia, and India, face higher hunger burdens (Fig. 8).

In Brazil, the rise in CA research coincided with the expansion of CA (zero-tillage practices), increased soybean production, and a decline in poverty rates (Fig. 9 a and b), highlighting the potential role of sustained research in advancing food security through CA adoption.

Discussion

Advancing tropical conservation agriculture amid unequal research leadership

The increasing body of research on CA in tropical regions reflects growing global recognition of sustainable land management as a strategy to address intertwined food security and environmental challenges. Notably, over half of the studies were published in the last decade, underscoring an urgent research response to escalating soil degradation, climate vulnerability, and agricultural underperformance in the tropics. This trend mirrors global policy agendas, particularly the UN Sustainable Development Goals on zero hunger and land restoration (Goals 2 and 15)[6].

Despite the notable progress in the research output, the findings reveal persistent reliance on expertise and institutional dominance from temperate regions. Brazil stands as a clear exception, emerging as a global leader in tropical CA, backed by sustained public investment, institutional prominence, and scientific capacity[23,24]. However, contributions from other tropical countries remain limited across institutional leadership, research funding, and authorship. This imbalance reflects the dominance of top-ranked institutions and funding agencies based in temperate zones. While early pioneers such as Rattan Lal shaped the field, the recent diversification of authors suggests an evolving and dynamic research landscape. Notably, the USA accounted for the most collaborative links, with temperate-tropical partnerships far outweighing intra-tropical networks. The scarcity of South-South collaboration represents a missed opportunity for horizontal knowledge exchange among countries facing comparable agroecological pressures. Therefore, strengthening tropical research leadership and regional collaboration is essential in collaboratively developing CA strategies that are ecologically relevant, socially inclusive, and better tailored to local contexts.

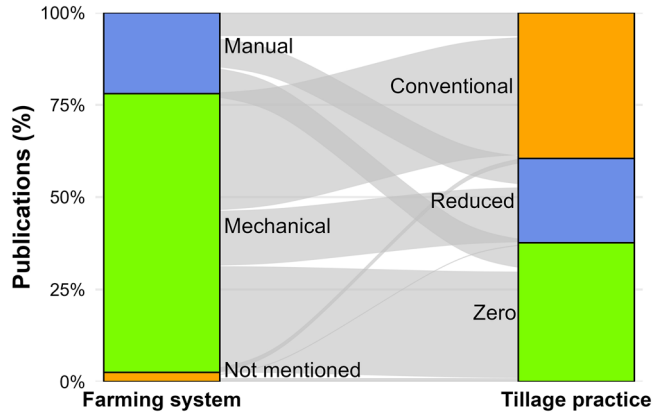


Fig. 7 | Sankey diagram showing the interconnections between the farming systems and tillage practices in the tropics. The height of the boxes and the thickness of the gray flows are proportional to the number of publications represented.

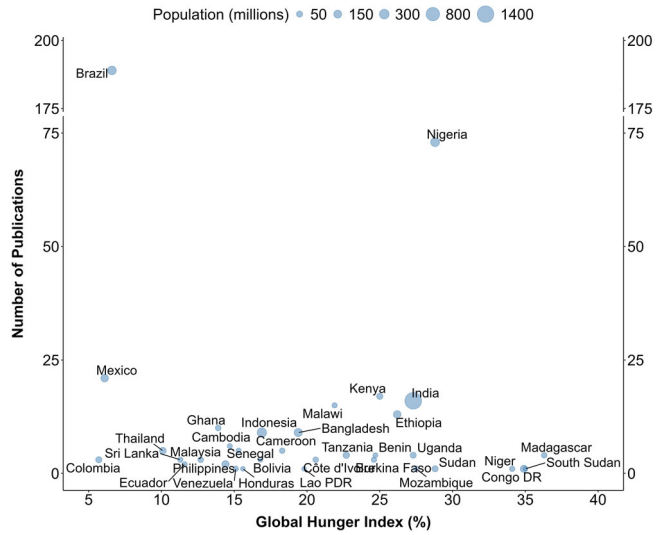


Fig. 8 | Number of publications on conservation agriculture and the hunger index across the tropics.

Assessment of soil health under conservation agriculture in the tropics

Monitoring soil health response to CA in the tropics is still a challenge. The predominant focus on chemical and physical soil properties in CA research reflects a legacy of evaluating soil fertility primarily through the lens of plant nutrition and fertilizer management - a perspective rooted in the Green Revolution era[25,26]. During this time, soil health was largely equated with nutrient availability and physical support for crops, which explains the emphasis on parameters such as organic carbon, nitrogen, phosphorus, pH, and bulk density. Globally, this reflects an evolving understanding of soil functionality and a shift in terminology, from soil fertility to quality, and currently soil health[27]. The gradual rise in biological assessments after the year 2000 is encouraging (Fig. 4a), signaling a growing recognition of soil biology as an integral dimension of soil science and ecosystem sustainability. This shift coincided with the establishment of the Soil Health Institute and increased strategic engagement from research institutions, which helped create global awareness and advance the conceptualization of soil health as an integrative framework encompassing physical, chemical, and biological attributes[28,29]. In the tropics, the Soil Bioanalysis technology (SoilBio) integrates soil enzymes, such as β-glucosidase and arylsulfatase, to assess soil health status[30]. The SoilBio framework establishes interpretive classes linked to crop yield and soil organic carbon, enabling standardized sampling procedures and the routine incorporation of biological indicators into commercial soil health analyses[31]. More recently, the Brazilian Soil Health Partnership was established, coordinated by the Center for Carbon Research in Agriculture (CCARBON) of the University of São Paulo - a multidisciplinary initiative that advances soil carbon research, climate-smart agriculture, and integrated soil health monitoring to support sustainable land management[17,32]. These advancements link soil health not only to agricultural productivity but also recognize that the well-being of humans, animals, and the environment is interconnected - one health concept[33–35].

Despite growing interest in soil health, few studies have assessed chemical, physical, and biological soil health indicators in response to CA (in the same study), and even fewer have applied structured frameworks to index soil health. This highlights a gap in the comprehensive assessment of CA effects on the multifaceted soil functions. In temperate regions, particularly in the United States of America (USA), frameworks such as the Soil Management Assessment Framework (SMAF) and the Comprehensive Assessment of Soil Health (CASH) have been successfully applied to monitor soil health response to land use and management, allowing a holistic soil health assessment in response to agricultural practices[36–40]. More recently, the Soil Health Assessment Protocol and Evaluation (SHAPE) framework, which also incorporates inherent soil properties and weather variables[41,42], has been developed but remains unexplored in tropical regions. Adapting and implementing such integrated soil health indices is crucial to advancing holistic evaluations and supporting tropical-specific agricultural management strategies.

Developing an effective site-specific soil health assessment framework for the tropics is challenging, but possible. A major constraint to developing soil health assessment frameworks or decision-support tools in the tropics is the limited availability of harmonized datasets that adequately capture the high heterogeneity of tropical soils and agroecological conditions[41]. Many legacy soil datasets are not properly georeferenced, limiting their usefulness for spatial assessments and model development, while the measurement of soil health indicators is often time-consuming and costly, resulting in fragmented datasets. Addressing these challenges will require expanding soil monitoring networks, improving access to soil information systems, harmonizing soil analytical protocols, and leveraging digital soil assessment approaches[43,44]. The development of frameworks such as SHAPE will depend on large, well-distributed soil datasets that capture the inherent heterogeneity of edaphic, climatic, and production system attributes[41]. These datasets can be integrated with emerging digital soil databases and remote sensing products to support the development of robust, data-driven soil health indices tailored to tropical environments.

The broad soil orders and textures represented in CA research underscore the diverse agroecological range of tropical regions. However, the concentration of studies in Oxisols, Alfisols, and Ultisols - inherent characteristics of weathered tropical soils - suggests a focus on areas most vulnerable to degradation, where CA can offer substantial benefits[45,46]. Although underrepresented in our study, soil groups such as Entisols and Vertisols remain significant in tropical production systems[47,48]. A key research limitation identified in this review is the predominant focus on topsoil layers, which limits the ability to capture the deeper benefits of CA on soil functions. While surface layers are important for immediate nutrient exchange, biotic habitat, and early root growth, subsoil horizons play pivotal roles. This is especially critical in Oxisols, Alfisols, and Ultisols, which were the most studied soil orders and are known for their deep profiles that can extend up to 200 cm[49–51]. These deep layers offer substantial potential for storing water, improving structural development, and sequestering carbon beyond the reach of rapid turnover processes in surface layers[52,53]. The neglect of these deeper horizons may therefore underestimate the cumulative benefits of CA, particularly under tropical conditions characterized by intense rainfall, drought cycles, and being more prone to climate change actions.

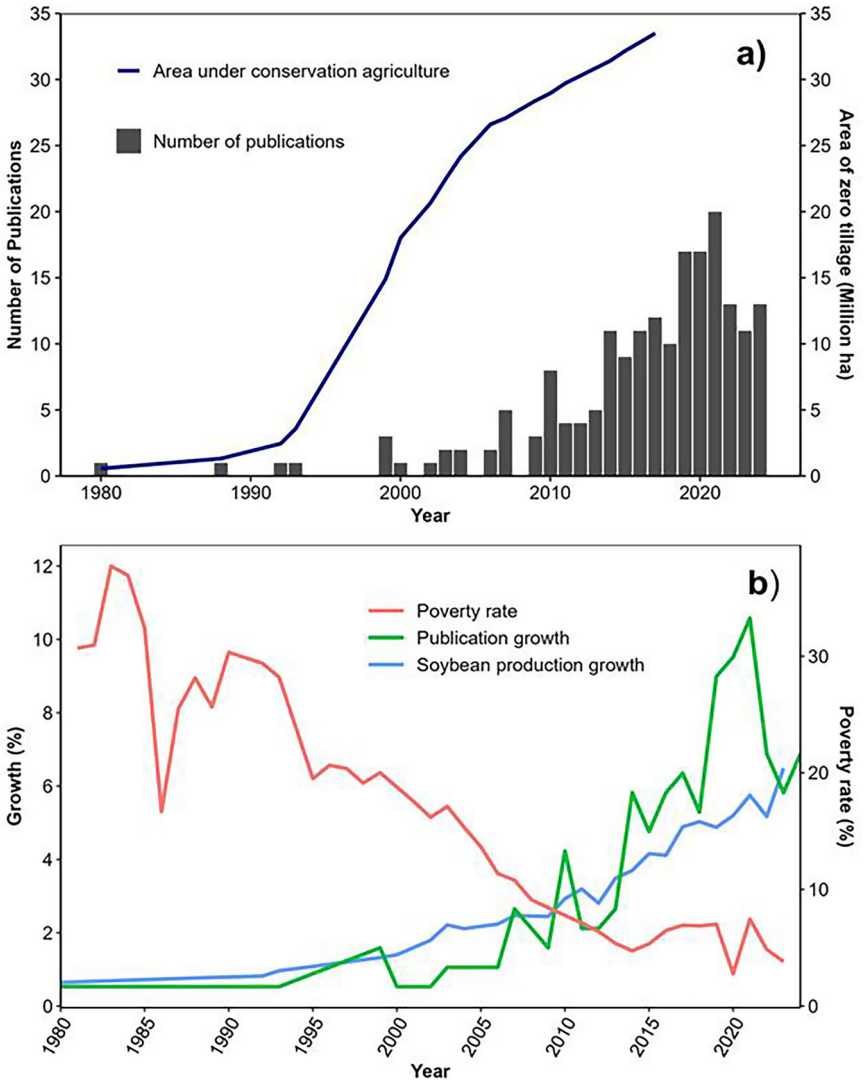


Fig. 9 | Conservation agriculture and soil health research outcomes. Number of research publications on conservation agriculture and the adoption of conservation tillage (**a**) and soybean production and poverty rate (**b**) within Brazil.

Management diversity in tropical conservation agriculture systems

The distribution of farming systems indicates that most CA studies in the tropics were conducted under mechanized production systems. The prominence of mechanized systems likely reflects the high operational requirements of CA practices, particularly zero tillage, which often depend on specialized equipment for herbicide-based weed control and row crop establishment[3]. Mechanization enhances efficiency and, therefore, dominates researchers' managed experiments and scalability in commercial production systems[54]. However, this pattern contrasts with the prevailing realities of tropical agriculture, where farmers rely on manual operations or animal traction, which can limit the adoption of CA practices that require multiple sequential operations[55]. This imbalance highlights a critical gap in the current research landscape and underscores the need to expand CA research to integrate mechanization within smallholder contexts that underpin food production in the tropical regions.

The diverse range of crop and management components integrated into tropical CA systems reflects the adaptability of CA principles across different farming contexts. The prominence of pastures and cover crops underscores their crucial role in maintaining soil cover, enhancing organic matter inputs, and stimulating biological activity[15,56]. The prominence of maize and soybean as cash crops reflects their widespread adoption by producers and strong demand for food, feed, and industrial uses, making them key drivers of food security and economic viability[1,57]. Maize serves as a major dietary staple and source of daily calories in many tropical regions, while soybean provides an important supply of plant-based protein and edible oil for both human consumption and livestock feed[58]. Smallholder farmers depend on these crops for their livelihood, whereas commercial systems increasingly utilize them for industrial processing, biofuel production, and export markets. Moreover, the soil management benefits derived from double cropping of maize and soybean systems, such as residue retention, increased microbial diversity, improved nutrient cycling, and higher yields, provide support for the broader adoption of CA across tropical farming systems[59,60].

The integration of more regionally relevant and indigenous crops into CA systems is essential for enhancing tropical farming systems. For example, there was low representation of root and tuber crops, despite their critical importance in tropical food systems. Crops such as cassava and yams are staple foods for millions of people, underscoring the food security role[61]. Interestingly, the top 15 global producers of cassava roots, including Nigeria, the Democratic Republic of Congo, and Thailand, are located within tropical regions[62]. Cassava is climate-resilient, having the adaptive capacity to tolerate high temperatures and variable precipitation due to climate change better than cereals[61,63]. However, the production of root and tuber crops is often characterized by intensive soil disturbance, high nutrient extraction, and susceptibility to erosion, which collectively exacerbate soil degradation and undermine long-term productivity[64–67]. These attributes highlight the opportunity to improve both soil health and local crop production through conservation-oriented approaches.

Furthermore, expanding research to include a broader range of crops, including fruits, vegetables, and indigenous crops, is essential to create more diversified and resilient food systems[68–70]. Such diversification not only strengthens ecological sustainability but also improves the economic viability and social acceptability of CA practices among smallholder farmers, who often rely on mixed cropping systems for income and sustenance.

Brazil leads the global tropics in conservation agriculture for soil health and food security

Brazil has emerged as a global leader in sustainable land management, driven by deliberate and sustained investment in agricultural science. The transformation of the once marginal Cerrado region reflects decades of institutional commitment to tropical agricultural innovation. A key milestone was the establishment of Brazilian Agricultural Research Corporation (EMBRAPA - Empresa Brasileira de Pesquisa Agropecuária) in the early 1970s, currently the largest agricultural research corporation in the tropics, which catalyzed national research engagement and the development of regionally adapted technologies[71].

Strategic public investments in education, research, and soil health management turned the Cerrado into one of the world's most productive agricultural frontiers[72,73]. This scientific revolution underpinned the high adoption of CA (Fig. 9a), positioning Brazil as the second-largest global adopter of CA practices after USA[4]. The limitation of this study to publications in English language may have underestimated the scientific contributions from tropical regions, where Portuguese, Spanish, and French are widely used in scientific communication. At the same time, publications in Portuguese that consider local contexts played important roles in facilitating national dissemination and adaptation of CA technologies in Brazil[74–76].

The impacts of this agricultural transformation profoundly increased food production (Fig. 9b). From a net food importer in the 1970s, Brazil is now the top global exporter of major crops such as soybeans, maize, coffee, and animal products. Soybean production alone increased by over 127 million between 1994 to 2023, contributing USD 6.72 billion to the national economy[62]. This expansion could be linked to CA-based practices and complemented by research on nutrient management and breeding[15,77,78]. The 85% increase in CA adoption between 2006 and 2017, reaching 33 million hectares, drives significant yield gains[79]. Besides high investment in agricultural research and adoption of CA, this increase in food production in Brazil can also be extensively associated with public policies, including agricultural credit programs, technical assistance and extension services, and export-oriented agricultural policies that promoted technology adoption, farm modernization, and productivity growth.

The case of Brazil demonstrates that scaling science-based CA in tropical regions can strengthen soil health. While our analysis does not establish a direct causal relationship, our findings highlight the potential contribution of sustained scientific research to enhancing food production and reducing poverty. Brazil's leadership supports food systems in low-, middle-, and high-income countries[80], reinforcing its role in advancing global food and economic security. These outcomes underscore the importance of aligning scientific research, policy support, and farmer adoption to deliver broad societal benefits.

As tropical regions face intensifying food insecurity (Fig. 8) consequent to land degradation, climate change, and burgeoning population, adapting Brazil's CA model, through localized research, South-South cooperation, and context-specific innovations, will be essential for achieving the Sustainable Development Goals (SDG 1, 2, and 15). This model offers not only agronomic resilience but also pathways for inclusive development rooted in indigenous knowledge and regional collaboration.

Methods

This study employed a two-pronged approach involving systematic review and bibliometric analysis (Fig. 1) to elucidate research trends on soil health measured in response to conservation agricultural practices across the tropical region.

Literature search strategy

A comprehensive literature search was conducted using the Web of Science Core Collection and Scopus databases. The search targeted article titles, abstracts, and keywords using a combination of terms related to CA, soil health, and tropical regions. Terms that are related to the core principles of CA were considered and included conservation agriculture, no-till, zero tillage, cover cropping, crop rotation, intercropping, and residue management. Soil health is the target variable, which terms include: soil health, soil fertility, soil quality, soil biological, soil chemical, and soil physical. The geographic focus was refined using tropics and a curated list of tropical countries. Countries with only small tropical regions, such as China and Australia, were excluded to avoid disproportionate search results. Boolean operators and wildcard symbols were used to capture variations in terminology.

The preliminary search was limited to peer-reviewed journal articles in English language, with the exclusion of publications in Portuguese (166), French (26), Spanish (16), Chinese (8), and German (5). Gray literature, such as theses, conference papers, and book chapters, was also excluded. The final search, completed in December 2024, returned 1935 records from Scopus and 1915 from Web of Science, exported in BibTeX and CSV formats.

Screening of articles followed the PRISMA protocol[81]. Records were compiled and de-duplicated using Zotero software. Studies solely on modeling and questionnaires were excluded after title and abstract screening. Hence, only studies presenting original field data on the effects of CA practices on soil properties were included, while reviews, modeling-only studies, and meta-analyses were excluded. To ensure accurate representation of research studies that assessed soil health under CA in the tropics, a rigorous article selection process involving manual screening was conducted. Further filtering ensured studies included a CA practice compared to a conventional control and explicitly measured soil health indicators (physical, chemical, and biological). Geographical relevance was verified by study coordinates or detailed location descriptions, cross-checked against tropical boundaries of latitudes 23° N and S. To avoid excluding studies conducted in tropical agroecosystems (e.g., the Brazilian Cerrado and parts of Mexico) whose reported coordinates slightly exceeded these limits, the boundary was marginally extended to 24.5° N and S.

Data extraction

The final screening identified 542 study sites from 474 publications on CA, spanning the period from 1974 to 2024. Full texts of the selected papers were reviewed, and data were manually extracted into a structured Microsoft Excel spreadsheet. Each entry of the Excel database included: (i) a unique identifier designated for each paper; (ii) publication metadata (authors, title, year, DOI, abstract); (iii) geographic information (coordinates and country). Articles without coordinates were derived from the described study sites using Google Maps; (iv) details of crop types used in the experiment. Crops were grouped by functional categories: cover crops, cereals, legumes, roots and tubers, oil and fiber crops, and sugar crops (primarily sugarcane). (v) inherent soil properties, including United States Department of Agriculture soil taxonomy and texture classifications. For studies from Brazil, the Brazilian Soil Classification System (SiBCS) was converted to USDA categories[82]. Textural data (sand, silt, clay) were extracted from studies, and where not reported, the textural fractions were obtained from SoiGrids 2.0 using the coordinates[83]. (vi) soil health indicators, defined as dynamic properties responsive to soil management practices[84]. Indicators were categorized into physical, chemical, and biological[7] and modified (Supplementary Table S5). Related properties were aggregated for harmonization across studies.

Relevant online data were compiled to determine the relationship between scientific research publications and key development indicators in the tropics, including the Global Hunger Index[85], country population[86], CA adoption[87], crop production[62], and poverty rates[88].

Data analysis

Data management, descriptive analyses, and mapping were conducted using R software[89]. Bibliometric analysis utilized the Bibliometrix R package and its Biblioshiny web interface[90]. Data from Scopus (csv format) were imported and pre-processed in the Biblioshiny web interface to standardize institutional names and author affiliations. Quantitative analyses also included trends in publication output, author nationality, keyword frequency, and collaboration networks.

Geographic distributions of CA studies were visualized using study coordinates, mapped with R packages such as sf and rnaturalearth. Trend and comparative figures were created using ggplot2, while intersections of soil health indicator use were illustrated with VennDiagram.

Data availability

The articles reviewed for this paper are being considered by the team to develop other studies. Data also used are: Global Hunger Index[85], country population[86], CA adoption[87], crop production[62], and poverty rates[88].

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Author contributions

OHO: Writing – original draft, Writing – review and editing, Data extraction, Validation, Methodology, Investigation, Formal analysis, Data curation, Software, Conceptualization, Resources. MRN: Writing – review and editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

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Competing interests

The authors declare no competing interests.

Additional information

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